



Original article

Factors Associated With the Changes in Smoking and Electronic cigarette use in Adolescents During the COVID-19 Pandemic: A Longitudinal Analysis



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A B S T R A C T

Purpose: Explore the factors associated with the changes in smoking and electronic cigarette (e-cigarette) use and susceptibility among adolescents during the COVID-19 pandemic.

Methods: We surveyed a cohort of students (7th–12th grade) from private schools in Guatemala. Baseline (May–September 2019) was conducted in-person and follow-up (June–November 2020) online during the lockdown. Separate Generalized Estimating Equations logistic models regressed current smoking ($n = 3,729$), current e-cigarette use ($n = 3,729$), smoking susceptibility among never-smokers ($n = 2,596$), and susceptibility to e-cigarette use among never-users ($n = 1,597$) on online ad exposure, visiting stores, social network smoking/e-cigarette use, substance use (alcohol, marihuana, and cigarette or e-cigarette), perceived harm of using cigarettes/e-cigarettes, socio-demographic characteristics, and survey wave. Interactions were assessed between time and ad exposures, friends smoking and e-cigarette use.

Results: Frequency of store visits, exposure to online ads, and the use of cigarette and e-cigarette lowered at follow-up. Online e-cigarette ads, having family and friends who smoke, and current e-cigarette use increased the likelihood of being a current smoker. Frequent exposure to online e-cigarette ads, having family who use e-cigarettes, and being a current smoker were associated with higher likelihood of current e-cigarette use. Exposure to either online ads or having friends that smoke or use e-cigarettes, increased susceptibility to using either product. Interaction results showed that high exposure to online e-cigarette ads overtime increased the susceptibility to use e-cigarettes.

Discussion: Exposure to online ads and friends and family cigarette and e-cigarette use increased adolescent consumption and susceptibility during the pandemic.

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IMPLICATIONS AND
CONTRIBUTION

In a sample of Latin American adolescents, these findings suggest that smoking and e-cigarette use and susceptibility to use either product is associated with ad exposures and the use of cigarettes and e-cigarettes among friends and family, even during social distancing and lockdown periods.

Conflicts of interest: All the authors declare no conflict of interests.

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Tobacco use continues to be the leading cause of preventable disease and deaths worldwide [1]. Even though cigarette smoking has steadily declined in some countries [2], adolescents and young adults are increasingly using electronic cigarettes

(e-cigarettes) [3]. E-cigarette use has even surpassed smoking in recent years among adolescents in many countries [3]. However, the onset of the COVID-19 pandemic was associated with declines in tobacco product use, including e-cigarettes [4]. Increased parental supervision, health concerns associated with the COVID-19 transmission, decreased product access, and reduced social gatherings were some of the factors associated to this decline [5].

Social influence from peers plays an important role in adolescent behavior, even more than the one from their parents [6]. In some groups, smoking and e-cigarette use are behaviors that provide access to and reinforce belonging in a peer group [7]. Furthermore, often peers are the main source for obtaining tobacco products [8]. The more adolescents have friends who smoke, the higher their risk for smoking [9], and continuing to smoke through adulthood [10]. However, with schools being closed due to the COVID-19 pandemic, adolescents spent less time with their peers and more with their parents, limiting their opportunities to acquire tobacco products. Consequently, the opportunities of social-related and peer-related consumption decreased, as shown by a meta-analytical study that found that in most cases, cigarette and e-cigarettes consumption decreased during the pandemic [11]. However, most data come from the United States and Europe, and two countries in Asia (China and Pakistan) leaving a knowledge gap on what happened with consumption elsewhere, including Latin America.

COVID-19 lockdown measures also limited the time spent in outdoor activities, and increased adolescents' screen time. A meta-analysis yield a mean increase in screen time of 52% compared to before the pandemic, being higher among adolescents from ages 12 to 18 years [12]. More screen time can lead to more exposure to tobacco and e-cigarette content in streaming services [13] and social media [14]. Before the pandemic, adolescents and young adults already encountered around 40% of cigarette and e-cigarette ads online [14]. Point-of-sale (POS) ads are another primary advertising channel and are associated with increases in smoking and smoking susceptibility among adolescents [15]. Adolescents' exposures to online and POS ads can promote positive perceptions of tobacco products and could spark interest in and use of cigarettes [15] and e-cigarettes [16]. In Guatemala, tobacco advertising during children programming hours and showing tobacco consumption by human models, public figures, athletes, or cartoons are not allowed in TV, radio, and outdoor advertising. Therefore, the POS [17] is a very important marketing channel, however, with the pandemic exposure at the POS decreased due to lockdowns. Though, no studies have evaluated the changes in the exposure to tobacco products ads before and during the pandemic.

Guatemala is a middle-income country with weak tobacco control and no e-cigarettes regulations. The last nationally representative school-based survey conducted in 2015 found that 15.7% of 13–15-year-old adolescents currently smoked and 5.6% currently used e-cigarettes [18]. A 2020 school-based survey of a convenience sample of Guatemalan adolescents found a higher prevalence of using e-cigarettes (27.7%) than cigarettes (8.7%) [19]. However, it is unknown how the COVID-19 lockdown could have influenced product consumption. During COVID-19 Guatemala implemented diverse social distancing measures, such as only allowing one person per family to do grocery shopping to avoid overcrowding, school closures, online teaching, and banning social gatherings.

Given that peer influence and ad exposure (online and at the POS) are the two factors that on their own can enable adolescent consumption of cigarettes and e-cigarettes, we aim to explore the factors associated to the changes in smoking and e-cigarette use and susceptibility among adolescents in Guatemala during the COVID-19 pandemic and the possible interactions among exposure to ads and peer influence use of cigarettes and e-cigarettes over time. We hypothesize (1) that during COVID the susceptibility to and current use of either product will be lower, (2) that exposure to advertising and having friends and family that use either product will be associated with higher susceptibility and use even during the COVID pandemic, and (3) that the advertising exposure and peer influence will interact over time increasing the strength of the association for susceptibility to and current use of cigarettes and e-cigarettes.

Methods

Study population

We analyzed data from an open cohort of students attending private schools in Guatemala City in two different time points. At baseline, students from 10 private schools from seventh grade through high school (ages 13–17 years) were included. Surveys were conducted in person from May–September 2019 (baseline) and online from June–November of 2020 (follow-up). Due to the pandemic, data collection in one school was not possible at follow up. Passive parental consent and students' consent was requested before completing the survey participation was voluntary and there was no compensation offered. More information regarding the cohort can be found elsewhere [20]. The survey included questions translated and, when necessary, adapted from the Population Assessment of Tobacco and Health study and the International Tobacco Control Youth survey [21], focusing on risk factors for smoking and e-cigarette use [22]. A total of 3,875 students answered the survey, after excluding those with missing values ($n = 146$), a total of 3,729 observations were analyzed (2,123 baseline and 1,606 at follow up). The protocol was approved by the institutional review board at the Central American Institute of Nutrition in Guatemala City.

Dependent variables

Current smoker and e-cigarette use were assessed with the questions “*In the past 30 days, how many days did you [smoke/use an e-cigarette]?*” (0 = “Nonsmoker/none-cigarette user” and 1 = “from 1 day to every day”). Susceptibility to smoking and e-cigarette use were assessed by asking “*Do you think at any time during the next 12 months, you will smoke/use an e-cigarette?*” and “*If one of your best friends offered you a [cigarette/e-cigarette], would you smoke/use it?*” (1 = “Definitely not” to 4 = “Definitely yes”). Those who responded with anything other than “definitely not” were classified as susceptible [23].

Independent variables

Exposure to ads. To assess ads exposure, the frequency of visiting stores was collected by asking “*In the past 30 days, how often have you visited convenience stores?*” (0 = “never”, 1 = “sometimes”, and 2 = “frequently and very frequently”). Predictive validity of this question has been proven in other studies [24]. Exposure to online marketing was assessed with the questions: “*In the past*

30 days, when browsing the internet or social networks (Instagram, Facebook, Snapchat, Twitter, and YouTube) how often have you seen advertisements for [regular cigarettes(pack)/e-cigarettes]? Answers were categorized into 0 = “never”, 1 = “sometimes”, and 2 = “frequently and very frequently” [25].

Social network use of cigarettes and electronic cigarettes

Family and friends smoking, and e-cigarette use was assessed by asking “If any family member living in their household [smokes/use e-cigarettes] (0 = no, 1 = yes)”, and “Of your five best friends, how many [smoke cigarettes/use e-cigarettes] (recoded: 0 = none, 1 = one or more).

Substance use

Participants were asked about their alcohol consumption as “In the past 30 days, how many days did you have at least one drink or one glass of an alcoholic drink? (recoded: 0 = none, 1 = one or more)” and marihuana use as “When was the last time that you smoked marihuana?” (recoded as: 0 = never smoked, 1 = ever use, due to low percentages of more frequent use).

Perceived harm

Perceptions regarding the harmful effects of both products were assessed with the questions “Assuming that you [smoke/use e-cigarettes] for the rest of your life, how likely would it be for you to get a serious illness from [smoking/using e-cigarettes]?” (0 = “I don’t know/it won’t happen/unlikely”, 1 = “likely/very likely”, and 3 = “surely”), and “How likely would it be for a person to become addicted to [cigarettes/e-cigarettes]?” (0 = “I don’t know, it won’t happen, unlikely, and likely”; 1 = “very likely and surely”) [26].

Sociodemographic

Covariates included sex (0 = female, 1 = male), school grade (1 = seventh–eighth, ninth, 2 = 10th, and 3 = 11th–12th), and highest educational attainment by either parent (1 = high-school or less, 2 = University, and 3 = don’t know).

Analysis

For the descriptive analysis we compared the sample characteristics at baseline and follow-up using Pearson chi-square for categorical and *t*-test for continuous variables. We fitted two pairs of logistic Generalized Estimating Equations. The first two models used the entire sample when regressing current smoking or e-cigarette use, modeled separately, on ad exposure, social network smoking, and e-cigarette use, alcohol drinking, marihuana use, current use of cigarettes or e-cigarettes (i.e., current e-cigarette use, included as independent variable on the smoking model and vice versa), perceived risk from smoking or e-cigarette (i.e., perceived harm of cigarette use was included in the smoking model, and perceived harm of e-cigarette use on the e-cigarette model), sociodemographic characteristics, and time (baseline vs. follow-up). The second pair of models restricted the sample to either never smokers (*n* = 2,596) or never e-cigarette users (*n* = 1,597) when regressing susceptibility to use either product, modeled separately, on the same set of covariates used in the first two models. We excluded from the e-cigarette

susceptibility model current smokers and marihuana users because there were no students in the nonsusceptible to e-cigarette use category. Interaction terms were added to adjusted models to assess whether time (i.e., pre-COVID baseline vs. COVID follow-up) moderated the effects of each of the following variables on tobacco product susceptibility and current use: ad exposure (online and store visits) and smoking and e-cigarette use among friends. After estimating each fully adjusted model, we added interaction terms one at the time and evaluated their statistical significance. For variables with three categories (online ads and store visits), we evaluated the global F-test for the block of interaction terms to evaluate statistical significance. When interactions were statistically significant, we re-ran the fully adjusted model after stratifying the data by survey wave. To evaluate the potential impact of excluding observations due to missing data, we re-estimated each fully adjusted model using multiple imputation with chained equations to impute missing values. The magnitude, direction, and statistical significance of the results from these sensitivity analyses (see [Appendix, Tables A1–A4](#)) were consistent with the complete case analyses, so are not discussed in the results. All analyses were conducted using Stata v.17.

Results

The frequency of visiting stores and exposure to online cigarette and e-cigarette ads was lower at follow-up (during COVID) compared to baseline (pre-COVID) across all three analytic samples (i.e., total sample, never smokers, and never e-cigarette users; *p* < .001). Similarly, the percentage of those who reported having family or friends who smoked or used e-cigarettes was lower at follow-up than baseline across all samples. Current use of cigarettes (10.2% vs. 3.1%) and e-cigarettes (31.3% vs. 14.1%) was also lower at follow-up, as it was the susceptibility to smoke (55.4% vs. 50.1%) and use e-cigarettes (72.6% vs. 63.4%) ([Table 1](#)).

In adjusted models, students in the follow-up during COVID had a higher likelihood of being susceptible to smoking (adjusted odds ratio [AOR] = 1.38, 95% (confidence intervals [CI]) [1.17, 1.61]), as did those who visited stores frequently or very frequently (AOR = 1.58, 95% CI [1.23, 2.03]), reported being exposed to online e-cigarette ads sometimes (AOR = 1.66, 95% CI [1.34, 2.06]), and frequently or very frequently (AOR = 2.79, 95% CI [1.95, 3.99]), and those who currently used e-cigarettes (AOR = 1.40, 95% CI [1.05, 1.87]). The use of cigarettes among social networks also increased smoking susceptibility ([Table 2](#)). Lower likelihood of being susceptible to smoking was observed on those who believed that people could get addicted to cigarettes (AOR = 0.76, 95% CI [0.63, 0.91]) ([Table 2](#)).

In adjusted models, students in the follow-up during COVID had a lower likelihood of being a current smoker (AOR = 0.41, 95% CI [0.28, 0.59]). Follow-up students had a higher likelihood of being current smokers if they had family members (AOR = 1.85 95% CI [1.32, 2.58]) or friends who smoke (AOR = 3.22, 95% CI [2.10, 4.92]), and currently used e-cigarettes (AOR = 4.98, 95% CI [3.45, 7.21]) ([Table 2](#)).

Visiting stores frequently or very frequently (AOR = 1.83, 95% CI [1.19, 2.82]), exposure online cigarette ads (AOR = 1.94, 95% CI [1.33, 2.82]), and having friends who use e-cigarettes (AOR = 1.68, 95% CI [1.16, 2.42]) increased the odds of being susceptible to using e-cigarettes at follow-up. While being male (AOR = 0.75, 95% CI [0.59, 0.95]), reporting that it is surely (AOR = 0.57, 95% CI [0.39, 0.82]) to develop a disease from using e-cigarettes or that

Table 1

Descriptive characteristics of the total sample (n = 3,729), never smokers (n = 2,596), and never e-cigarette users (n = 1,597) by wave

	Total sample			Never smokers			Never e-cigarette users		
	Baseline (n = 2,123)	Follow-up (n = 1,606)		Baseline (n = 1,383)	Follow-up (n = 1,213)		Baseline (n = 860)	Follow-up (n = 737)	
Sex, %									
Female	50.2	53.4		53.6	55.2		53.8	54.8	
Male	49.8	46.6		46.4	44.8		46.2	45.2	
School grade, %									
7th–8th grade	27.7	23.0	*	36.5	26.8	**	43.7	30.3	**
9th grade	29.7	32.7		31.5	37.5		27.1	38.8	
10th grade	26.9	27.6		20.6	24.2		18.7	21.3	
11th–12th grade	15.7	16.7		11.4	11.5		10.5	9.6	
Parents education, %									
High-school or less	18.7	17.9	*	18.4	18.1		17.3	17.6	
University	73.0	70.6		72.3	69.8		72.1	69.6	
Don't know	8.3	11.5		9.3	12.1		10.6	12.8	
Frequency of visiting stores, %									
Never	13.3	32.5	**	14.7	34.8	**	15.7	38.4	**
Sometimes	54.5	50.9		55.9	50.8		56.9	48.4	
Frequently–very frequently	32.2	16.6		29.4	14.4		27.4	13.2	
Exposure to online ads for cigarettes, %									
Never	33.5	50.8	**	38.1	54.6	**	39.2	57.8	**
Sometimes	55.2	44.6		52.2	41.5		51.2	38.8	
Frequently–very frequently	11.3	4.6		9.7	3.9		9.6	3.4	
Exposure to online ads for e-cigarettes, %									
Never	29.5	39.4	**	35.4	44.3	**	41.6	51.9	**
Sometimes	52.8	51.8		50.9	49.5		46.0	44.0	
Frequently–very frequently	17.7	8.8		13.7	6.2		12.4	4.1	
Family smoke, %	45.6	36.6	**	39.3	33.4	*	37.3	29.7	*
Friends smoke, %	58.8	47.1	**	47.9	40.7	**	41.5	35.7	*
Family uses e-cigarettes, %	24.3	18.6	**	18.9	15.3	*	10.4	10.2	
Friends use e-cigarettes, %	67.1	59.5	**	58.4	52.1	*	43.9	38.9	*
Current smoker, %	10.2	3.1	**	–	–		–	–	
Current e-cigarette user, %	31.3	14.1	**	16.1	6.5	**	–	–	
Alcohol use, %	46.9	38.5	**	31.3	29.8		20.1	18.5	
Marihuana use (ever), %	11.5	8.3	*	1.7	1.0		0.8	0.5	
Perceived harm from cigarette use (disease), %									
I don't know/it won't happen/unlikely	7.9	7.5		7.5	6.7		7.8	6.6	
Likely/very likely	53.4	55.7		51.8	55.2		48.4	52.7	
Surely	38.7	36.7		40.7	38.1		43.8	40.7	
Perceived addictiveness of cigarettes, %	71.2	67.7	*	74.8	68.8	*	74.8	67.8	*
Perceived harm from e-cigarette use (disease), %									
I don't know/it won't happen/unlikely	27.5	16.3	**	25.1	14.7	**	22.6	12.6	**
Likely/very likely	58.6	67.8		58.8	68.6		57.4	67.6	
Surely	13.9	15.9		16.1	16.7		20.0	19.8	
Perceived addictiveness of e-cigarettes, %	61.1	65.0	*	63.1	65.9		64.8	67.2	
Susceptibility to smoke, %	55.4	50.1	*	38.3	38.1		30.5	26.1	*
Susceptibility to use e-cigarettes, %	72.6	63.4	**	60.5	54.2	*	43.8	34.1	**

P values from chi-square test for categorical variables and t-test for continuous variables *p value <.050, **p-value <.001.

e-cigarettes can cause addiction (AOR = 0.71, 95% CI [0.56, 0.89]) lowered the odds of susceptibility to e-cigarette use (Table 3).

Adjusted models for current e-cigarette use yield lower likelihood of use among those at follow-up (AOR = 0.38, 95% CI [0.31, 0.46]) and those who had friends who smoked (AOR = 0.71, 95% CI [0.54, 0.93]). Greater likelihood of e-cigarette use was found for those who were exposed to online e-cigarette ads sometimes (AOR = 1.33, 95% CI [1.02, 1.74]) and frequently or very frequently (AOR = 1.84, 95% CI [1.20, 2.48]), were male (AOR = 1.36, 95% CI [1.11, 1.68]), had family (AOR = 2.84, 95% CI [2.25, 3.59]) or friends (AOR = 6.49, 95% CI [4.37, 8.91]) who use e-cigarettes, and those who currently smoked (AOR = 4.59, 95% CI [3.15, 6.68]) (Table 3).

When adding interaction terms to the models, only the interaction between wave (pre-COVID and during COVID) and exposure to online e-cigarette ads in the model for e-cigarette susceptibility was statistically significant ($p = .0234$) (Table 4). When stratified by wave, we found that at baseline, frequent

exposure to online e-cigarette ads increased the (AOR = 2.24 95% CI [1.21, 4.15]) of being susceptible to use e-cigarettes, compared to never being exposed. At follow-up, being exposed to online e-cigarette ads sometimes (AOR = 2.17 95% CI [1.39, 3.37]) and frequently (AOR = 4.62 95% CI [1.59, 13.39]) increased the odds of being susceptible to use e-cigarettes (Table 5).

Discussion

According to our findings, frequency of exposure to ads, peer use of cigarettes and e-cigarettes, and substance use decreased during the follow-up period (during COVID-19) compared to the baseline period (pre-COVID-19). This was expected due to lockdowns and school closures. The lockdown in Guatemala was not as strict as in other countries. In this country, the feasibility of going out was limited and with the social distancing measures in place the opportunities to visit grocery or convenience stores,

Table 2

Factors associated with smoking susceptibility and current smoking pre-COVID and during COVID

	Smoking susceptibility (n = 2,596)				Current smoker (n = 3,729)			
	OR	95% CI	AOR	95% CI	OR	95% CI	AOR	95% CI
Wave								
Pre-COVID	Ref.		Ref.		Ref.		Ref.	
During COVID	1.03	(0.89–1.17)	1.38**	(1.17–1.61)	0.31**	(0.24–0.41)	0.41**	(0.28–0.59)
Frequency of visiting stores								
Never	Ref.		Ref.		Ref.		Ref.	
Sometimes	1.33*	(1.11–1.59)	1.21	(0.98–1.48)	2.24**	(1.51–3.32)	1.14	(0.66–1.96)
Frequently-very frequently	1.79**	(1.44–2.23)	1.58**	(1.23–2.03)	3.60**	(2.38–5.44)	1.27	(0.72–2.26)
Exposure to online ads for cigarettes								
Never	Ref.		Ref.		Ref.		Ref.	
Sometimes	1.80**	(1.53–2.11)	1.25*	(1.01–2.07)	1.76**	(1.33–2.34)	1.13	(0.79–1.63)
Frequently-very frequently	2.01**	(1.47–2.75)	1.02	(0.69–1.52)	3.13**	(2.10–4.64)	1.13	(0.63–2.02)
Exposure to online ads for e-cigarettes								
Never	Ref.		Ref.		Ref.		Ref.	
Sometimes	2.12**	(1.79–2.50)	1.66**	(1.34–2.06)	1.83**	(1.31–2.57)	0.99	(0.64–1.51)
Frequently-very frequently	3.40**	(2.57–4.48)	2.79**	(1.95–3.99)	5.40**	(3.73–7.83)	1.79*	(1.07–3.00)
Sex								
Female	Ref.		Ref.		Ref.		Ref.	
Male	0.97	(0.81–1.15)	0.96	(0.79–1.15)	1.23	(0.94–1.61)	1.05	(0.75–1.47)
School grade								
7th–8th grade	Ref.		Ref.		Ref.		Ref.	
9th grade	1.03	(0.86–1.22)	0.88	(0.72–1.07)	1.85*	(1.22–2.79)	0.73	(0.42–1.26)
10th grade	0.99	(0.80–1.22)	0.71*	(0.56–0.89)	4.12**	(2.74–6.18)	1.13	(0.66–1.95)
11th–12th grade	0.73*	(0.55–0.96)	0.48**	(0.35–0.66)	4.19**	(2.70–6.49)	1.04	(0.58–1.86)
Parents education								
High-school or less	Ref.		Ref.		Ref.		Ref.	
University	1.14	(0.92–1.41)	1.13	(0.89–1.43)	0.87	(0.64–1.18)	0.87	(0.60–1.25)
Don't know	1.11	(0.81–1.52)	1.23	(0.88–1.72)	0.61	(0.36–1.03)	1.00	(0.52–1.94)
Family smoke								
No	Ref.		Ref.		Ref.		Ref.	
Yes	1.45**	(1.23–1.72)	1.26*	(1.05–1.52)	2.81**	(2.16–3.65)	1.85**	(1.32–2.58)
Friends smoke								
No	Ref.		Ref.		Ref.		Ref.	
Yes	1.64**	(1.40–1.92)	1.44**	(1.18–1.76)	6.29**	(4.40–9.01)	3.22**	(2.10–4.92)
Family uses e-cigarettes								
No	Ref.		Ref.				Ref.	
Yes	1.54**	(1.24–1.91)	1.02	(0.80–1.30)	2.59**	(2.00–3.34)	0.86	(0.61–1.21)
Friends use e-cigarettes								
No	Ref.		Ref.		Ref.		Ref.	
Yes	1.71**	(1.46–2.02)	1.11	(0.89–1.36)	4.76**	(3.24–7.00)	0.85	(0.52–1.38)
Current e-cigarette user								
No	Ref.		Ref.		Ref.		Ref.	
Yes	2.00**	(1.56–2.58)	1.40*	(1.05–1.87)	15.56**	(11.39–21.26)	4.98**	(3.45–7.21)
Alcohol use (past 30 days)								
No	Ref.		Ref.		Ref.		Ref.	
Yes	2.30**	(1.93–2.73)	1.99**	(1.63–2.42)	12.54**	(8.42–18.68)	3.27**	(2.02–5.29)
Marihuana use (past 12 months)								
No	Ref.		Ref.		Ref.		Ref.	
Yes	2.44*	(1.29–4.58)	1.44	(0.68–3.05)	13.83**	(10.33–18.53)	4.65**	(3.28–6.59)
Perceived harm from smoking (disease)								
I don't know/it won't happen/unlikely	Ref.		Ref.		Ref.		Ref.	
Likely/very likely	1.54*	(1.13–2.09)	1.67*	(1.20–2.32)	0.70	(0.47–1.05)	0.76	(0.47–1.23)
Surely	1.09	(0.79–1.49)	1.21	(0.86–1.69)	0.45**	(0.29–0.71)	0.60	(0.35–1.02)
Perceived addictiveness of smoking								
No	Ref.		Ref.		Ref.		Ref.	
Yes	0.81*	(0.68–0.96)	0.76*	(0.63–0.91)	0.62**	(0.49–0.81)	0.70*	(0.51–0.97)

P-values: *p-value < .050, **p-value < .001.

where cigarettes and e-cigarettes are advertised were reduced. Consequently, adolescents had less time to be out and were less exposed to ads at the POS, which is high in Guatemala [17]. Furthermore, schools changing to online classes likely limited the opportunity for adolescents to purchase or share cigarettes or e-cigarettes with peers.

Adolescent social networks are a key determinant in smoking initiation and use [27]. These networks have also been found to increase adolescent e-cigarettes use [28]. We found that having

family or friends who smoke or used e-cigarettes increased the odds of being a smoker or e-cigarette user. Interestingly, having friends who smoke or use e-cigarettes was consistently associated with higher susceptibility to using either product, contrary to family. As children grow, peers become more important in influencing each other's behavior. This could be related to the adolescent vulnerability to social influence as they try to fit in with their peers, the importance of seeking connections during this age, and the personal process of independence and

Table 3

Factors associated with e-cigarette susceptibility and current e-cigarette use pre-COVID and during COVID

	E-cigarette use susceptibility (n = 1,597)				Current e-cigarette user (n = 3,729)			
	OR	95% CI	AOR	95% CI	OR	95% CI	AOR	95% CI
Wave								
Pre-COVID	Ref.		Ref.		Ref.		Ref.	
During COVID	0.74**	(0.62–0.87)	0.70	(0.44–1.12)	0.43**	(0.38–0.49)	0.38**	(0.31–0.46)
Frequency visit stores								
Never	Ref.		Ref.		Ref.		Ref.	
Sometimes	1.60**	(1.27–2.02)	1.02	(0.68–1.51)	1.73**	(1.42–2.11)	0.93	(0.72–1.21)
Frequently-very frequently	2.50**	(1.87–3.34)	1.83*	(1.19–2.82)	2.60**	(2.09–3.24)	1.13	(0.84–1.53)
Exposure to online ads for cigarettes								
Never	Ref.		Ref.		Ref.		Ref.	
Sometimes	2.17**	(1.77–2.65)	1.94*	(1.33–2.82)	1.58**	(1.35–1.85)	0.91	(0.69–1.17)
Frequently-very frequently	2.49**	(1.63–3.78)	1.62	(0.84–3.11)	2.25**	(1.75–2.91)	0.67	(0.45–1.00)
Exposure to online ads for e-cigarettes								
Never	Ref.		Ref.		Ref.		Ref.	
Sometimes	2.15**	(1.75–2.63)	1.15	(0.79–1.67)	1.95**	(1.63–2.33)	1.33*	(1.02–1.74)
Frequently-very frequently	3.79**	(2.59–5.53)	1.75	(0.98–3.12)	3.83**	(3.05–4.82)	1.84*	(1.20–2.48)
Sex, %								
Female	Ref.		Ref.		Ref.		Ref.	
Male	0.75*	(0.60–0.94)	0.75*	(0.59–0.95)	1.21*	(1.02–1.42)	1.36*	(1.11–1.68)
School grade								
7th – 8th grade	Ref.		Ref.		Ref.		Ref.	
9th grade	1.02	(0.82–1.26)	0.97	(0.76–1.24)	1.96**	(1.58–2.41)	1.58*	(1.19–2.09)
10th grade	0.91	(0.70–1.19)	0.75	(0.55–1.01)	2.87**	(2.31–3.57)	1.69*	(1.25–2.26)
11th – 12th grade	0.84	(0.59–1.19)	0.71	(0.48–1.06)	3.22**	(2.53–4.09)	1.79**	(1.28–2.47)
Parents education								
High-school or less	Ref.		Ref.		Ref.		Ref.	
University	1.14	(0.87–1.49)	0.97	(0.72–1.29)	0.94	(0.78–1.14)	0.91	(0.70–1.18)
Don't know	1.02	(0.69–1.48)	0.98	(0.65–1.48)	0.59*	(0.43–0.81)	0.92	(0.61–1.39)
Family smoke								
No	Ref.		Ref.		Ref.		Ref.	
Yes	1.35*	(1.09–1.68)	1.06	(0.78–1.45)	1.78**	(1.52–2.07)	0.88	(0.71–1.08)
Friends smoke								
No	Ref.		Ref.		Ref.		Ref.	
Yes	1.40*	(1.14–1.72)	0.98	(0.68–1.42)	2.52**	(2.15–2.95)	0.71*	(0.54–0.93)
Family uses e-cigarettes								
No	Ref.		Ref.		Ref.		Ref.	
Yes	1.14	(0.82–1.60)	0.69	(0.42–1.15)	3.83**	(3.24–4.54)	2.84**	(2.25–3.59)
Friends use of e-cigarettes								
No	Ref.		Ref.		Ref.		Ref.	
Yes	1.92**	(1.57–2.34)	1.68*	(1.16–2.42)	9.67**	(7.52–12.42)	6.49**	(4.37–8.91)
Current smoker								
No	Ref.		Ref.		Ref.		Ref.	
Yes	-	-	-	-	13.29**	(9.98–17.69)	4.59**	(3.15–6.68)
Alcohol use								
No	Ref.		Ref.		Ref.		Ref.	
Yes	2.80**	(2.16–3.63)	2.69**	(2.03–3.56)	8.54**	(7.09–10.27)	4.27**	(3.46–5.29)
Marihuana use (last year)								
No	Ref.		Ref.		Ref.		Ref.	
Yes	-	-	-	-	7.02**	(5.55–8.88)	3.54**	(2.62–4.79)
Perceived harm (disease)								
I don't know/it won't happen/unlikely	Ref.		Ref.		Ref.		Ref.	
Likely/very likely	0.79	(0.61–1.02)	0.81	(0.61–1.07)	0.74**	(0.63–0.88)	1.11	(0.78–1.25)
Surely	0.53**	(0.38–0.72)	0.57*	(0.39–0.82)	0.52**	(0.43–0.66)	0.93	(0.56–1.13)
Perceived harm (addiction)								
No	Ref.		Ref.		Ref.		Ref.	
Yes	0.74*	(0.60–0.90)	0.71*	(0.56–0.89)	0.89	(0.77–1.03)	0.94	(0.77–1.15)

P values: *p value < .050, **p value < .001.

self-discovery [29]. During the pandemic, adolescents relied heavily on social media to maintain their connection with friends. A study on substance use, found that during this period, some reported using substances in virtual contexts with their friends, indicating that social substance use is not limited to face-to-face interactions [30]. According to our findings, even though adolescents were not able to meet with their peers face-to-face during the pandemic, the influence of friends remained significant with an unchanged strength of the association before and

during the pandemic. This finding supports that peer smoking remains one of the strongest and most consistent predictors of adolescent smoking and e-cigarette use.

Our results showed that using e-cigarettes was associated with increased odds of smoking and smoking susceptibility. Current smokers also had higher odds of current e-cigarette use. A strong relationship between e-cigarette use and subsequent smoking has been previously documented [31] and between smoking and subsequent e-cigarette use among adolescents [32].

Table 4

Results for the interaction for wave (pre-COVID and during COVID) and exposure to online ads for e-cigarettes over the susceptibility to use e-cigarettes

	OR	95% CI	p value
Exposure to online ads for e-cigarettes*Wave			
Never	Ref.		
Sometimes	1.92	(1.14–3.26)	<i>p</i> = .0234
Frequently–very frequently	3.16	(1.04–9.62)	

Our results point to a positive association between the two products and more research should be conducted in order to disentangle if e-cigarettes are a gateway into cigarette smoking [33] or whether there is a common liability pathway (i.e., shared risk factors) in the use of these products [34]. The fact that starting to consume either product can lead adolescents to consume the other is highly relevant since there is an increased probability for adolescents to become addicted to nicotine and continue smoking into adulthood. As expected, being sure that one can get a disease from using either product or that these products can get you addicted was associated to lower odds of being current user or susceptible to either product. Therefore, it is worth considering tailoring interventions to include information related to harm perception and addictiveness for tobacco products among adolescents.

We observed that frequent exposure to online e-cigarette ads was associated with higher likelihood of using e-cigarettes and cigarettes, as well as susceptibility to smoking. Similarly, susceptibility to smoking or using e-cigarettes increased in those who reported being exposed to online ads for cigarettes and e-cigarettes, respectively. Exposure to online ads has been significantly associated with smoking initiation [35], intention and current e-cigarette use, as well as lower e-cigarettes harm and addictiveness perceptions [36]. Additionally, the exposure to e-cigarette ads had been found to lower harm perceptions of smoking among children that have not used tobacco products [37]. This may indicate an association between e-cigarette ads, where usually these products are glamorized or advertised as healthy, and a possible renormalization with smoking.

Our results also yield that those who reported visiting stores were more susceptible to smoking and using e-cigarettes. This suggests that POS ads can influence adolescents even during lockdown. Evidence suggests that adolescents will still recall smoking and e-cigarette ads long after watching them and that this is linked to an increase in the risk of smoking or using e-cigarettes or being susceptible to use [38]. In Guatemala, tobacco and e-cigarette advertising, promotion, and sponsorship is for the most part unregulated and the tobacco industry has focused on POS as an advertising channel [17]. Therefore, policies to regulate ads (online and at the POS) are needed to mitigate the

effect of ads in future consumption and susceptibility among adolescents.

The prevalence of online ad exposures and visiting stores where tobacco products are advertised and sold decreased from pre-COVID to during COVID. However, the strength of association between these variables and susceptibility and use of cigarettes and e-cigarettes did not change over time. It is unclear whether the declines in ad exposure account for the decrease in use of both products at follow-up, and whether the declines in use led to less noticing of ads because of selective attention bias [39]. Furthermore, posting of tobacco product could have dropped due to lower use. Future research should try to capture posting patterns to better understand changes in exposure.

Our interaction term showed that the association between high exposure to online e-cigarette ads was more strongly associated with e-cigarette susceptibility during the pandemic compared to before. The fact that this increase was only found with e-cigarettes could be related to the more prevalent use of these products among adolescents [3], and more influencers (i.e., *people in social media who promotes products, services, or brands, and have a large number of followers*) promoting e-cigarettes than cigarettes. It has been observed that influencers who promote e-cigarettes do not restrict their content to adults and are followed by many adolescents [40]. The way e-cigarettes are advertised has been linked with never e-cigarette users perceiving them as cooler, more fun and enjoyable, and less harmful [41]. Ads are a powerful tool to promote e-cigarettes and it is arguably the most important reason behind their popularity among adolescents. Therefore, there's a need to reduce adolescents' ad exposure to prevent the e-cigarette use and susceptibility.

Our study has some limitations. Data were collected through self-report and there were no biomarkers collected, so data should be interpreted with caution. Given the COVID-19 lockdown, follow-up data were collected online at the student's home. Even though students had the option to skip any question for any reason, due to the nature of the lockdown, is possible that they were not alone in the room, possibly generating some bias as privacy could have been compromised. However, declines in use were also documented elsewhere [11] therefore, our results are in the range of expected. Follow-up data were collected during the COVID-19 lockdown. However, no COVID-specific variables (e.g., infection, perceived risks), that could have affected susceptibility to use either product was captured. Data were collected in private schools that mostly include middle-income and high-income students, limiting the generalizability of the results. The extent of any bias is unknown given the lack of other Guatemalan data. E-cigarette use may be higher among students with greater expendable income and access to new technologies, but it is unknown whether correlates of use would differ for other groups. To the best of our knowledge this is the

Table 5

Results from the exposure to online e-cigarette ads over the susceptibility to use e-cigarettes stratified by wave (pre-COVID and during COVID)

	Baseline			Follow-up		
	AOR	95% CI	p value	AOR	95% CI	p value
Exposure to online ads for e-cigarettes						
Never		Ref.			Ref.	
Sometimes	1.30	(0.87, 1.95)	.198	2.17	(1.39–3.37)	.001
Frequently–very frequently	2.24	(1.21, 4.15)	.010	4.62	(1.59–13.39)	.005

Models adjusted by ad exposure, social network smoking and e-cigarette use, alcohol drinking, marihuana use, current use of e-cigarettes, perceived risk of e-cigarettes, and sociodemographic characteristics.

first study that explored the factors associated to the changes in smoking and e-cigarette use and susceptibility among adolescents during the COVID-19 pandemic in a Latin American country.

In conclusion, our results suggest that during the COVID-19 pandemic the frequency of exposure to POS and online cigarette and e-cigarette advertising decreased, and so did social network smoking and e-cigarette use. However, despite the lockdown and social distancing measures, the impact that friends and ads have among adolescents prevailed, leading to increased susceptibility and use of both the products.

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Supplementary Data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.jadohealth.2023.11.399>.

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